Reflections and Impedance

If you've ever been on a boat in calm water, you might think the ocean is a very quiet place. If you've ever gone snorkelling, though, you may have noticed that's very much not the case. The ocean is *loud*! Why don't we hear it?

1. Reflection and Tranmission

We've seen that when a wave hits the end of a string or pipe, the reflection depends on what's happening at the end of the string/pipe. If the end is loose, the reflected wave comes back with the same sign as the wave we sent in. If the end is fixed, the reflected wave flips sign. But what happens if we attach two strings with different masses?

Let's start by assuming we have two strings tied together with different masses per unit length λ_m, and corresponding wave speeds v_l and v_r for the left part and the right part of the string. We send a wave in; what happens when it hits the boundary? We know if we send the wave in from the left, if the right string is very heavy, we get a negative reflection, because the right spring isn't going to move (much). If the right string is very light, it might as well not be there, so we'll get a positive reflection. If the right string is exactly the same as the left, the wave will just keep travelling happily because we just have a single uniform string. What about other values, though?

As usual, we have to figure out what happens if we zoom in to the boundary. If we have an incoming wave, we know we'll have a reflected wave, but some of the wave will keep moving to the right. For simplicity, we'll assume an incoming sine wave y_i:

$$y_i = A\sin(k_i x - \omega_i t) \tag{1}$$

We'll also have a reflected wave y_r:

$$y_r = R\sin(-k_r x - \omega_r t) \tag{2}$$

You might want to look at Equation 2 for a moment to convince yourself it's a left-moving sine wave. If we increase t we have to *decrease* x to keep the inside fixed, so the wave has to be moving left. Finally, we have a transmitted wave y_t

$$y_t = T\sin(-k_t x - \omega_t t) \tag{3}$$

We aren't allowed to break the string, so the height on the right must equal the height on the left at $x = 0$. That gives us

$$A\sin(-\omega_i t) + R\sin(-\omega_r t) = T\sin(-omega_t t) \tag{4}$$

This has to be true at any value of time t (because we can never break the string). The way to make this happen is to set $\omega_i = \omega_r = \omega_t = \omega$. Otherwise if we don't break the string for some time, we will break it at another time. Once we know the ω's are the same, we can divide through by $\sin(-\omega t)$ to get

$$A + R = T \tag{5}$$

That is, the amplitude of the incoming plus reflected wave must equal the amplitude of the outgoing wave. This is our first condition.

For the second condition, we know that if the acceleration ever goes to infinity at the boundary, the string will respond instantly to make the acceleration finite. If the strings have the same tension (which will be the case if we're pulling on a single string), then I can't have a kink in the string. Otherwise we get a finite force on an infinitessimal bit of string, and the acceleration goes to infinity. If I try to kink the string, it responds instantly, so we never actually get a kink. No kink means the *slopes* match on the right and the left. How does the slope depend on k? Well, higher k means more waves crammed into the same space, which means higher slope. So the *slope* of $\sin(kx)$ must look like k times the slope of sin[1]. The slope on the left is the sum of the incoming and reflected slopes, but because the reflected wave has been flipped around, its slope is negative. We get the slope on the left is

$$Ak_iZ - Rk_rZ = Tk_tZ \tag{6}$$

where Z is the slope of sin. We know that $v = \frac{\omega}{k}$ or $k = \frac{\omega}{v}$. Since everybody has the same frequency, and the incoming and reflected waves have the same speed (since they are on the same string) that $k_i = k_r$. We can divide out the factor of Z to get

$$(A - R)k_i = Tk_t \tag{7}$$

or

$$A - R = \frac{k_t}{k_i}T \tag{8}$$

We know that $k = \frac{\omega}{v}$ so we have

$$\frac{k_t}{k_i} = \frac{\omega/v_t}{\omega/v_i} = \frac{v_i}{v_t} \tag{9}$$

We plug Equation 9 into Equation 8 to get

$$A - R = \frac{v_i}{v_t}T \tag{10}$$

Lets define $\alpha \equiv \frac{v_i}{v_t}$ then our conditions are

$$A + R = T \tag{11}$$

$$A - R = \alpha T \tag{12}$$

If we add these equations together, the R's cancel, and we get

$$2A = (1 + \alpha)T \tag{13}$$

$$T = \frac{2}{1+\alpha}A = \frac{2v_t}{v_i + v_t}A \tag{14}$$

If we multiply the first equation by α and subtract, then we cancel out the T's, and get:

$$A(\alpha - 1) + R(\alpha + 1) = 0 \tag{15}$$

$$R = \frac{\alpha - 1}{\alpha + 1}A = \frac{v_t - v_i}{v_t + v_i}A \tag{16}$$

[1]If you've taken calculus, you probably know that the slope, or derivative, of sin is cos, but we don't actually need that. We just need to know that the slope of $\sin(kx)$ is k times the slope of sin.

Equations 14 and 16 tell us the amplitudes of the reflected and transmitted waves, in terms of the wave speed in the two parts of the string. If the string on the right is very heavy, then v_t goes to zero and the transmitted wave amplitude is 0 times the incoming amplitude, which is always 0. As the right string gets heavier and heavier, the corresponding wave gets smaller and smaller. Similarly, the reflected wave turns into $\frac{-v_i}{v+i} = -1$, so the reflection comes back negative. This is exactly what we expected, since a super-heavy string looks like a fixed boundary.

The behavior when the right string is super light is perhaps more interesting. Now we have v_i goes to zero because v_t is so much larger. Our transmitted amplitude is $2\frac{v_t}{v_t} = 2$, so the wave amplitude on the right string is twice as high as the incoming wave. The reflected wave has amplitude $\frac{v_t}{v_t} = 1$, so it comes back with the *same* sign as the incoming wave. We've seen this before at an free end, but now we know that has we made that free end by tying on a very light string, the transmitted wave has twice the height. Finally, what happens when the speeds are the same? The transmitted amplitude matches the incoming, and the reflected is zero because $v_t - v_i = 0$ if $v_t = v_i$. As expected, the wave just keeps going as if there wasn't a junction.

For finite but different values of v_t and v_i, the absolute value of the reflection decreases. If the speed on the right is lower, then the junction acts sort-of like a fixed end, and we get a negative reflection. The closer the two speeds are, the smaller the reflection. If the right string is lighter, the speed is higher, and we get a postive reflection. The transmitted amplitude is up to 2 if the right string has a higher speed, it's the same as the incoming when the speeds match, and goes to zero as the speed drops. You can see this in action on desmos here, where you can adjust the k in the left region, and the k in the right region (which I've labelled u) separately.

2. Discussion and Impedance

We've seen that for a string, if the wave speed changes, waves will get partially reflected and and partially transmitted. More generally, the quantity we actually care about is called the *impedance*, and is closely related to the sound speed. We'll discuss this more later, but for now note that the speed of sound in water is much higher than the speed of sound in air. Sound going from water to air is going from high speed to low speed, or high impedance to low impedance, so very little makes it through. The sound all gets reflected off the surface. The ocean may be a noisy place, but what happens in the ocean, stays in the ocean. Incidentally, your ear is designed to match air, so sounds still sound muffled underwater. If you want to record a sound underwater, you should use a hydrophone, not a microphone!